

Muscle Metabolism

For PC-II Students

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Objective of the session

- At the end of this session students will be able:-
 - ✓ Define Metabolism
 - ✓ List energy sources for muscle contraction in different metabolic states.
 - ✓ Discuss anaerobic glycolysis and manifestation of muscle cramp,
 - ✓ Elaborate lactate re- utilization(Cori cycle).
 - ✓ Explain Muscle glycogenesis and glycogenolysis,

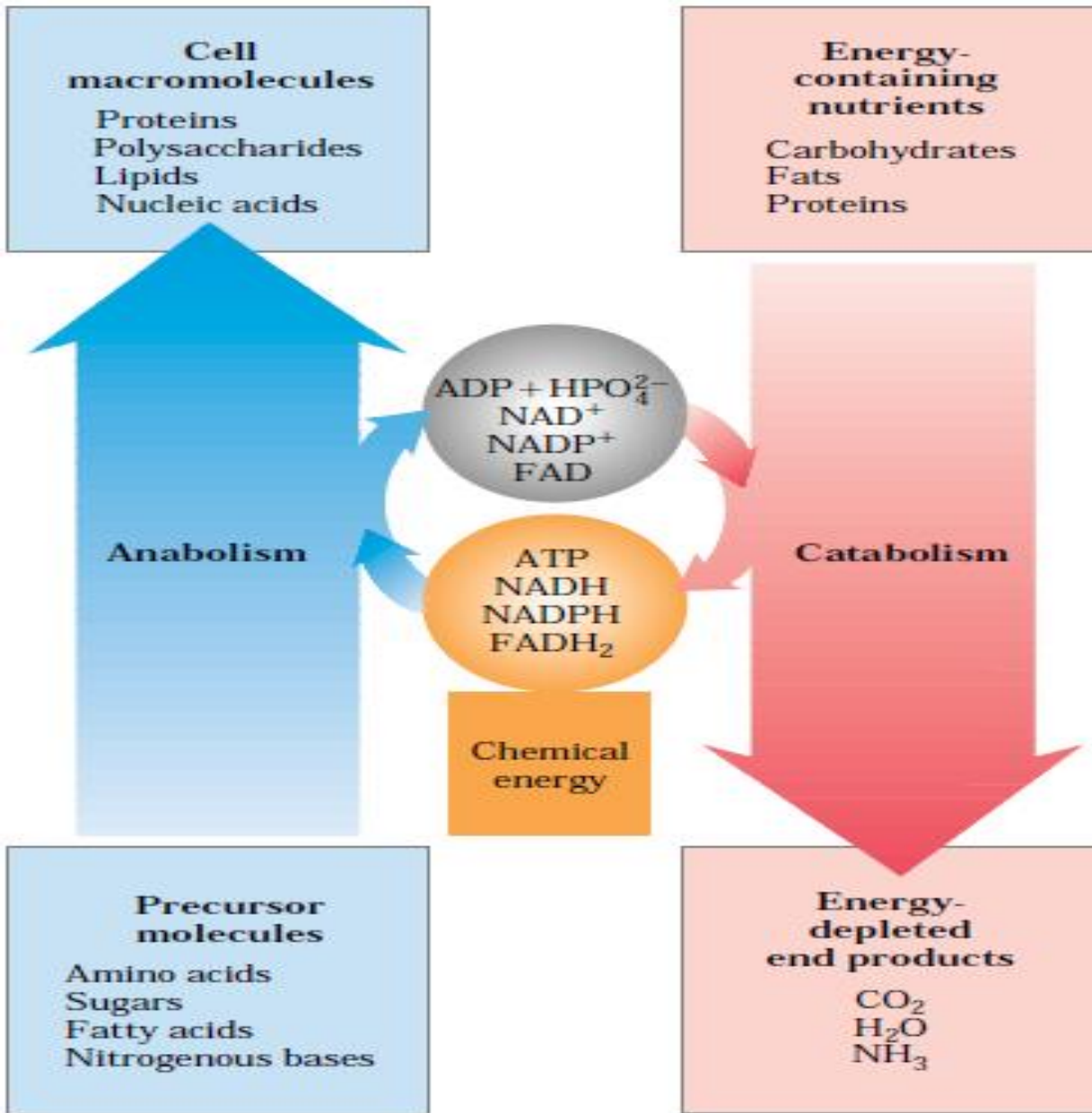
Metabolism

- **Metabolism**, the sum of all the chemical transformations taking place in a cell or organism, occurs through **metabolic pathways**.
- They have two phases
 - I. Catabolism** is the degradative phase of metabolism in which organic nutrient molecules (**carbohydrates, fats, and proteins**) are converted into smaller, simpler end products (such as lactic acid, CO₂, NH₃).
 - ✓ It release energy, conserved in the formation of ATP and reduced electron carriers (NADH, NADPH, and FADH₂); the rest is lost as heat(entropy)

Cont...

II. In anabolism phase, also called biosynthesis, small, simple precursors are built up into larger and more complex molecules, including **lipids, polysaccharides, proteins, and nucleic acids**.

- ✓ Anabolic reactions require an **input of energy, generally in the form of the phosphoryl group transfer potential of ATP and the reducing power of NADH, NADPH, and FADH₂**



- Catabolic pathways deliver chemical energy in the form of **ATP, NADH, NADPH, and FADH_2** .
- These energy carriers are used in **anabolic pathways** to convert small precursor molecules into cell macromolecules.

FIG. Energy relationships between catabolic and anabolic pathways.

Muscle Metabolism

- Energy containing Nutrients (sources of energy in heterotrophic organism) includes:-
 - ✓ **Carbohydrate**
 - ✓ **Lipids**
 - ✓ **Proteins**
- **Sources of energy in Muscle**
 - ✓ Blood glucose
 - ✓ Muscle glycogen
 - ✓ Fatty acids
 - ✓ Ketone bodies
 - ✓ Phosphocreatine

Cont...

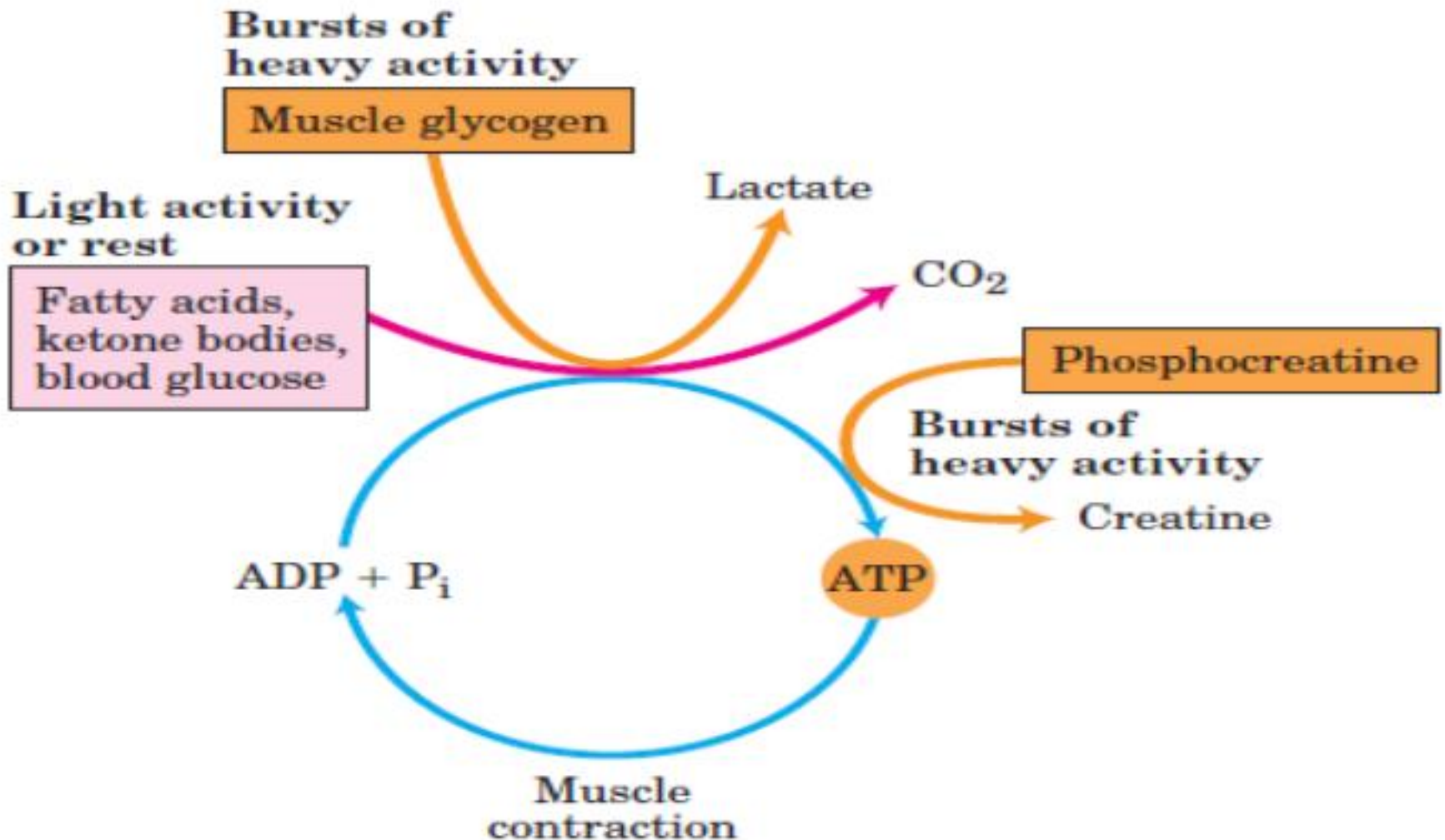


FIGURE 23-17 Energy sources for muscle contraction. Different fuels are used for ATP synthesis during bursts of heavy activity and during light activity or rest. Phosphocreatine can rapidly supply ATP.

Cont...

- The **myocytes** of skeletal muscle is specialized to generate ATP as the immediate source of energy for contraction.
- Skeletal muscles may work at their maximum capacity for a short time or prolonged work as in marathon or extended physical labor
- There are **two general classes of muscle tissue**, which differ in physiological role and fuel utilization
 - ✓ **Slow-twitch muscle**
 - ✓ **Fast-twitch muscle**

Cont...

- I. **Slow-twitch muscle**, also called red muscle, provides relatively low tension but is highly resistant to fatigue
- It produces ATP by the relatively slow but steady process of oxidative phosphorylation
 - Red muscle is very **rich in mitochondria** and is served by very dense networks of **blood vessels**, which bring the oxygen essential to ATP production
 - It is the **cytochromes in mitochondria** and the **hemoglobin in blood** that give the tissue its characteristic red color

Cont...

II. Fast-twitch muscle, or **white muscle**, has fewer mitochondria than red muscle and is less well supplied with blood vessels, but it can develop greater tension, and do so faster.

- White muscle is quicker to fatigue, because when active, it uses ATP faster than it can replace it
- There is a genetic component to the proportion of red and white muscle in any individual; with training, the endurance of fast-twitch muscle can be improved.

Cont...

- Skeletal muscle can use **free fatty acids, ketone bodies, or glucose** as fuel, depending on the degree of muscular activity
- **In resting muscle**, the primary fuels are **free fatty acids** from adipose tissue and **ketone bodies** from the liver.
 - ✓ These are oxidized and degraded to yield acetyl-CoA, which enters the **citric acid cycle** for oxidation to CO₂.
 - ✓ The ensuing transfer of electrons to O₂ provides the energy for ATP synthesis by oxidative phosphorylation.

Cont...

- **Moderately active muscle** uses blood glucose in addition to fatty acids and ketone bodies.
 - ✓ The glucose is phosphorylated, then degraded by glycolysis to pyruvate, which is converted to acetyl-CoA and oxidized via the citric acid cycle and oxidative phosphorylation

Cont...

- In maximally active fast-twitch muscles(vigorous exercises), the demand for ATP is so great that;
 - ✓ the blood flow cannot provide O₂, so ATP cannot synthesis by oxidative phosphorylation.
 - ✓ Under these conditions, stored muscle glycogen is broken down to lactate by fermentation(anaerobic respiration).

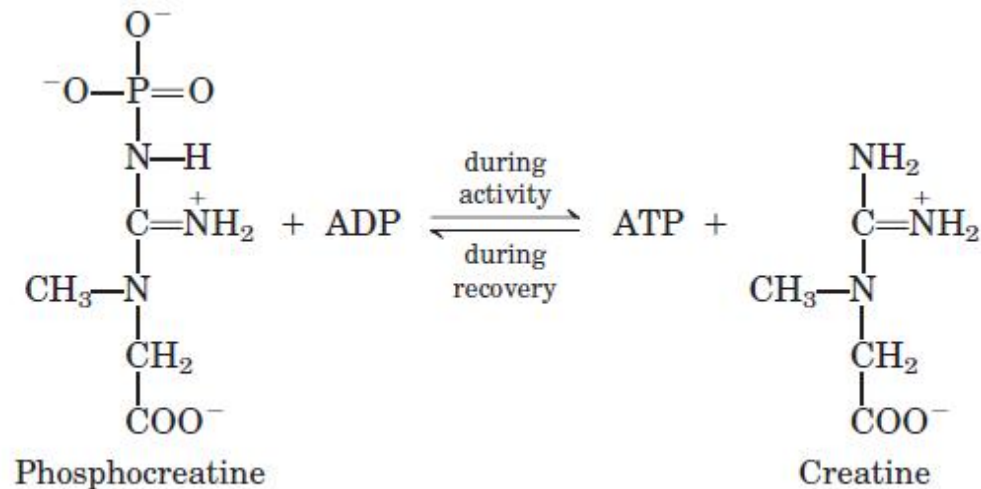
Cont...

- ✓ The use of blood glucose and muscle glycogen as fuels for muscular activity is greatly enhanced by the secretion of **epinephrine**, which stimulates both the release of **glucose** from **liver glycogen** and the breakdown of **glycogen in muscle tissue**.

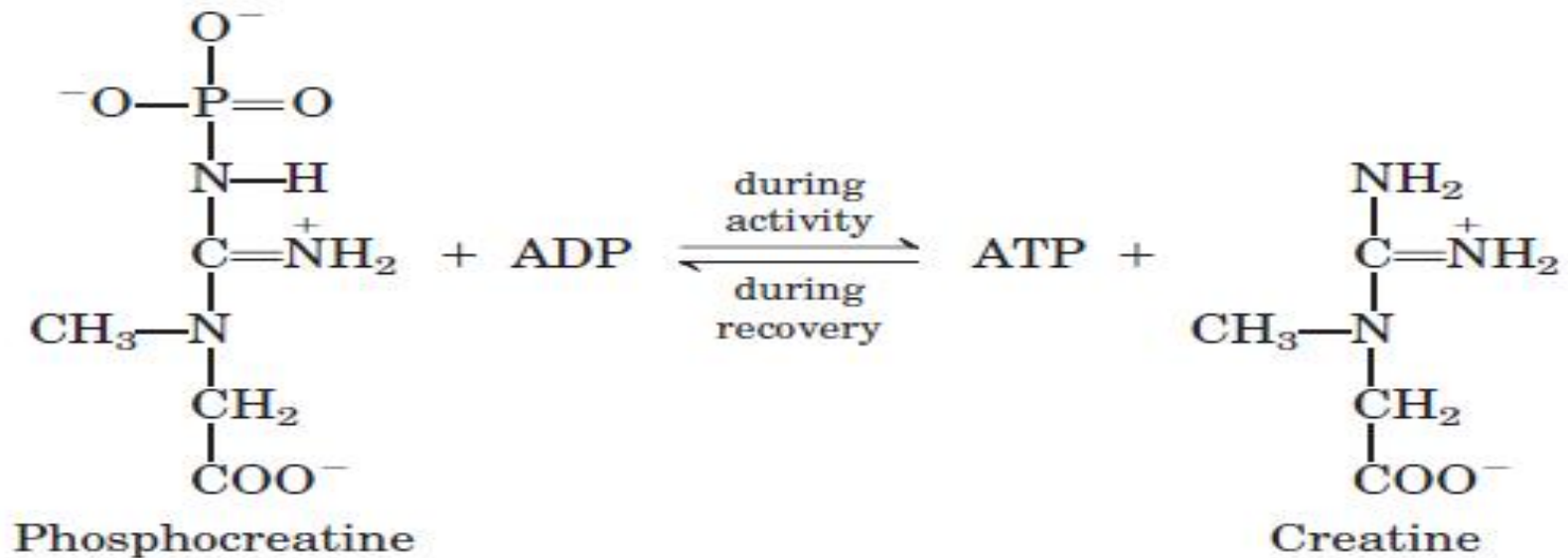
Cont...

- ✓ Moreover, the accumulation of lactate and consequent decrease in pH in maximally active muscles reduces their efficiency.
- ✓ Skeletal muscle, however, contains another source of ATP, in the form of phosphocreatine (10 to 30 mM), which can rapidly regenerate ATP from ADP by the

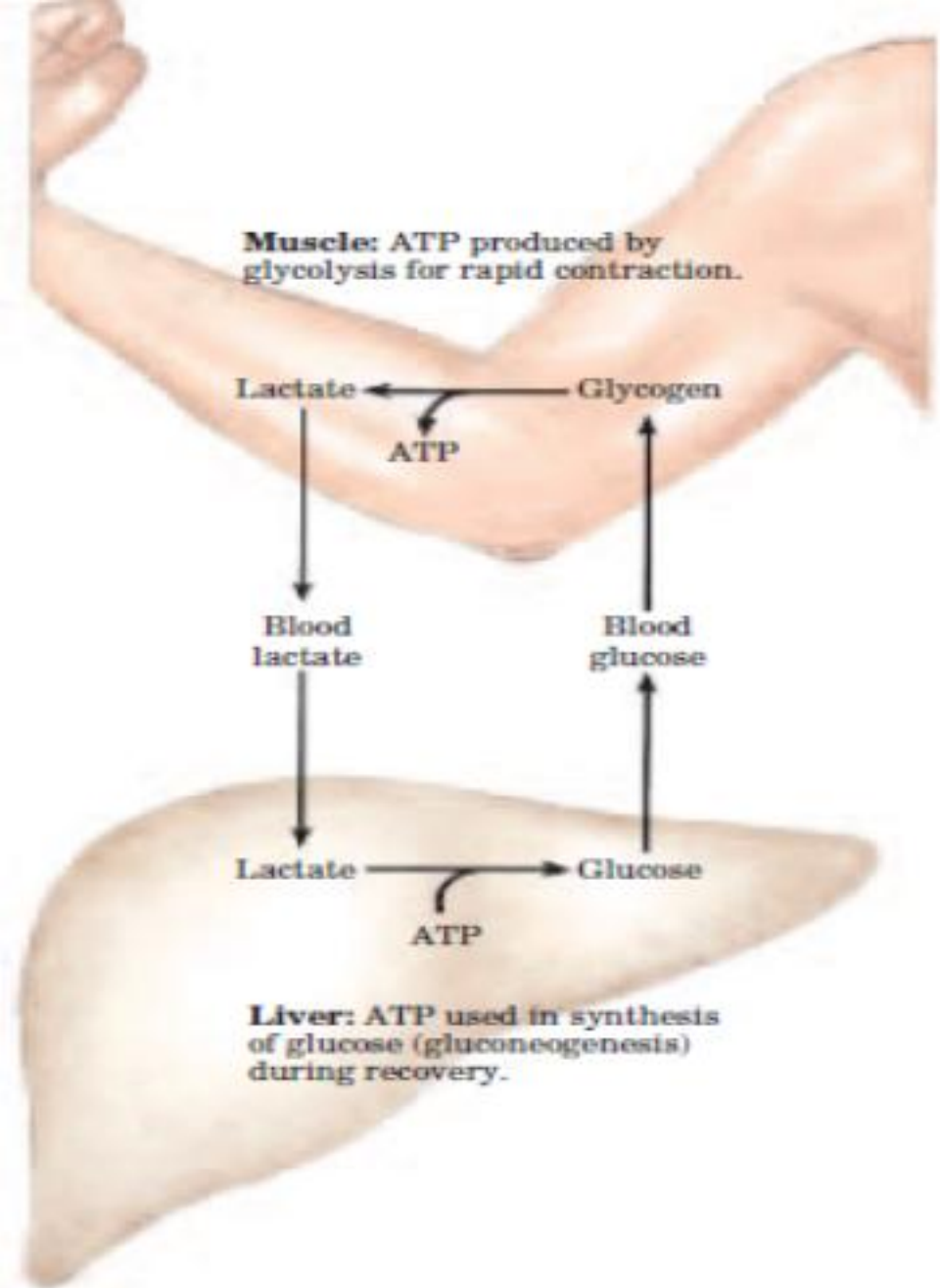
creatine kinase reaction



Cont...



- ✓ During periods of active contraction and glycolysis, this reaction proceeds predominantly in the direction of ATP synthesis; during recovery from exertion,
- ✓ the same enzyme resynthesizes phosphocreatine from creatine at the expense of ATP.



- ✓ Extremely active muscles use glycogen as energy source, generating lactate via glycolysis.
- ✓ During recovery, some of this lactate is transported to the liver and converted to glucose via gluconeogenesis.
- ✓ This glucose is released to the blood and returned to the muscles to replenish their glycogen stores.
- ✓ The overall pathway (glucose → lactate → glucose) constitutes the Cori cycle.

FIGURE 23-18 Metabolic cooperation between skeletal muscle and the liver. Extremely active muscles use glycogen as energy source.

Cont...

- ✓ **Heart muscle** differs from skeletal muscle in that it is continuously active in a regular rhythm of contraction and relaxation, and it has a completely aerobic metabolism at all times
- ✓ Mitochondria are much more abundant in heart muscle than in skeletal muscle, making up almost half the volume of the cells
- ✓ The heart uses as its fuel mainly **free fatty acids**, but also some **glucose** and **ketone bodies** taken up from the blood; these fuels are **oxidized via the citric acid cycle** and oxidative phosphorylation to generate ATP.
- ✓ Like skeletal muscle, heart muscle does not store lipids or glycogen in large amounts

Cont...

- ✓ It does have small amounts of reserve energy in the form of phosphocreatine, enough for a few seconds of contraction.
- ✓ Because the heart is normally aerobic and obtains its energy from oxidative phosphorylation,
- ✓ the failure of O₂ to reach a portion of the heart muscle when the blood vessels are **blocked by lipid deposits** (atherosclerosis) or blood clots (coronary thrombosis) can cause that region of the heart muscle to die.
- ✓ This is what happens in **myocardial infarction**, more commonly known as a **heart attack**.

Cont...

Muscle Metabolism:- includes

I) Carbohydrate metabolism

- ✓ Glycolysis(Embden -Meyerhoff pathway)
- ✓ Glycogen synthesis(glycogenesis)
- ✓ Glycogen degradation(Glycogenolysis)

II) Lipid metabolism

- ✓ Fatty acid oxidation(Beta-oxidation)
- ✓ Biosynthesis of Cholesterol(but minimal)

III) Amino acid/ protein metabolism

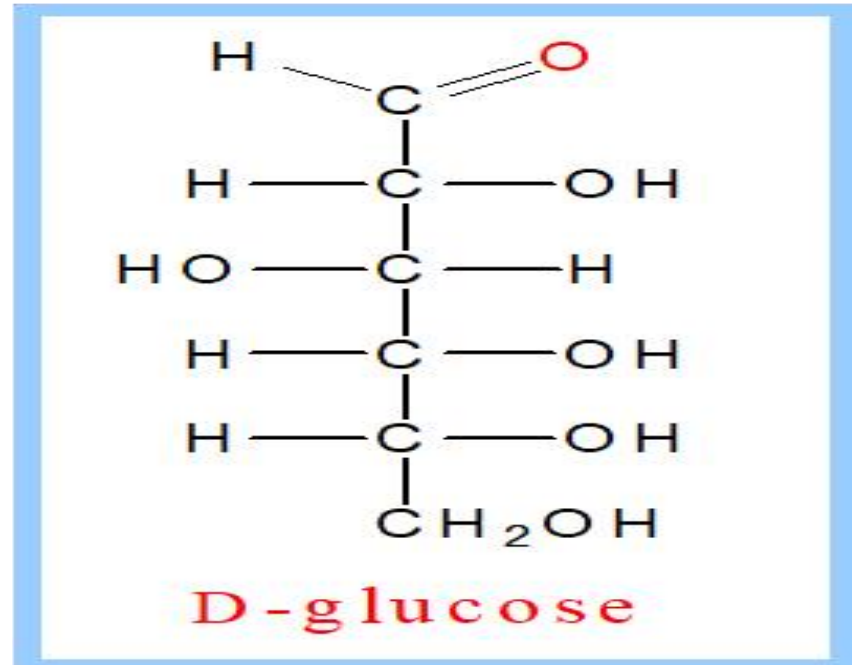
- ✓ Transamination etc.

IV) Intermediary metabolism

- ✓ Krebs's cycle
- ✓ Oxidative phosphorylation

Glycolysis (**Embden -Meyerhoff pathway**)

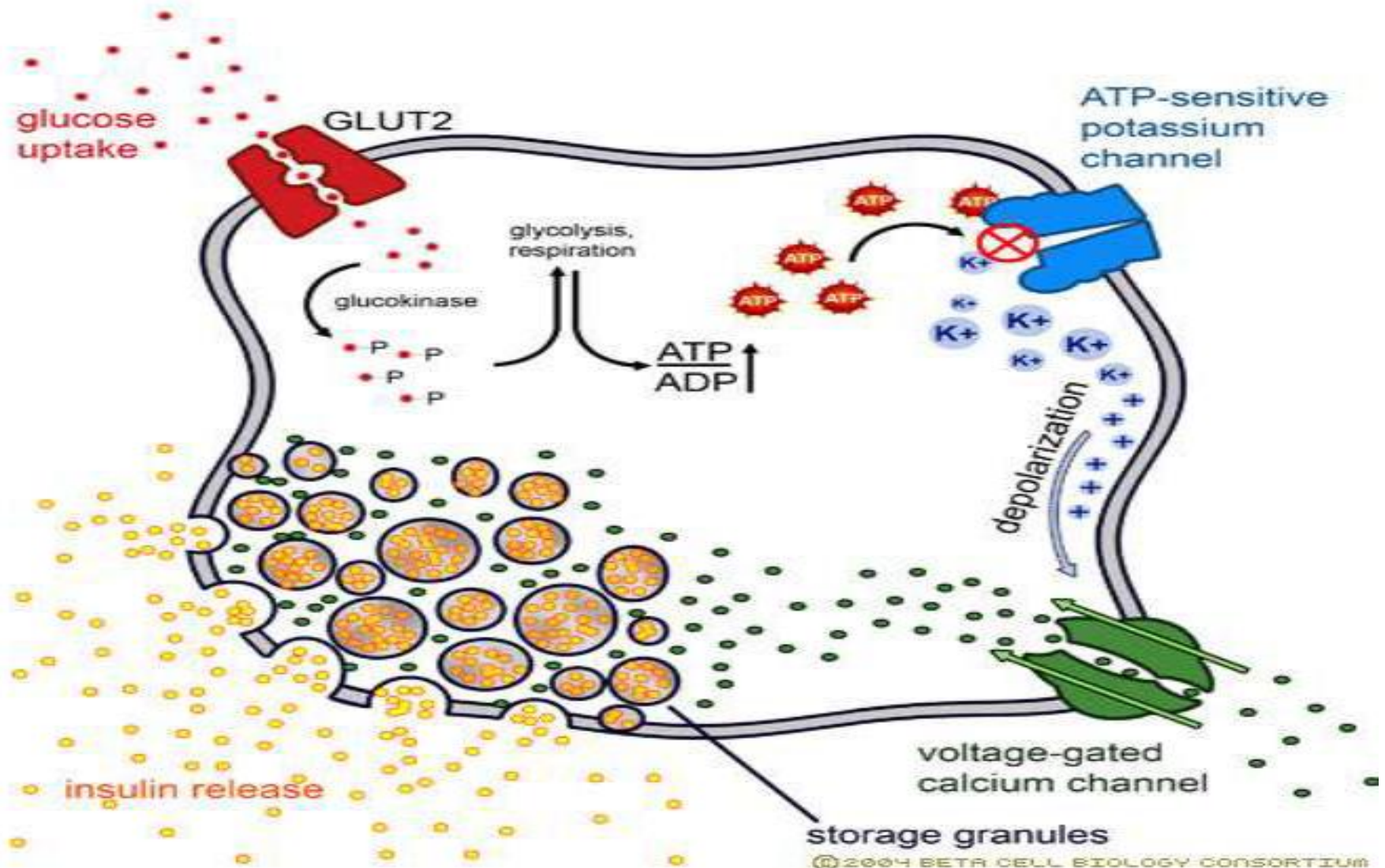
- **Glycolysis:-** is oxidation of **glucose to pyruvate** and lactate
 - ✓ It occurs virtually **in all tissues**.
- ❖ **what is glucose?**
 - ✓ Other name of glucose **Dextrose or Grape Sugar**
 - ✓ It is one of the monosaccharide carbohydrate used as primary source of energy in most tissue, which contain 6 carbon, structural look like the following.



Source and fate of glucose in the cells

- **Source of glucose** in the cells could be:-
 - ✓ **From dietary source(blood glucose)**
 - ✓ From glycogen degradation (Glycogenolysis)
 - ✓ **From non-carbohydrate molecules (Gluconeogenesis)**
 - ✓ From others monosaccharides (fructose and galactose)
- **Blood glucose** homeostasis is regulated mainly by two hormones: **Insulin and Glucagon**
- **Insulin** secreted when blood glucose is high.
 - ✓ Controls transport of glucose from blood to **muscle and fat cells**

INSULIN SECRETION



Fates of glucose in the cells

➤ **Fates of Glucose** in the cells could be:-

I. Oxidative (catabolic) fates

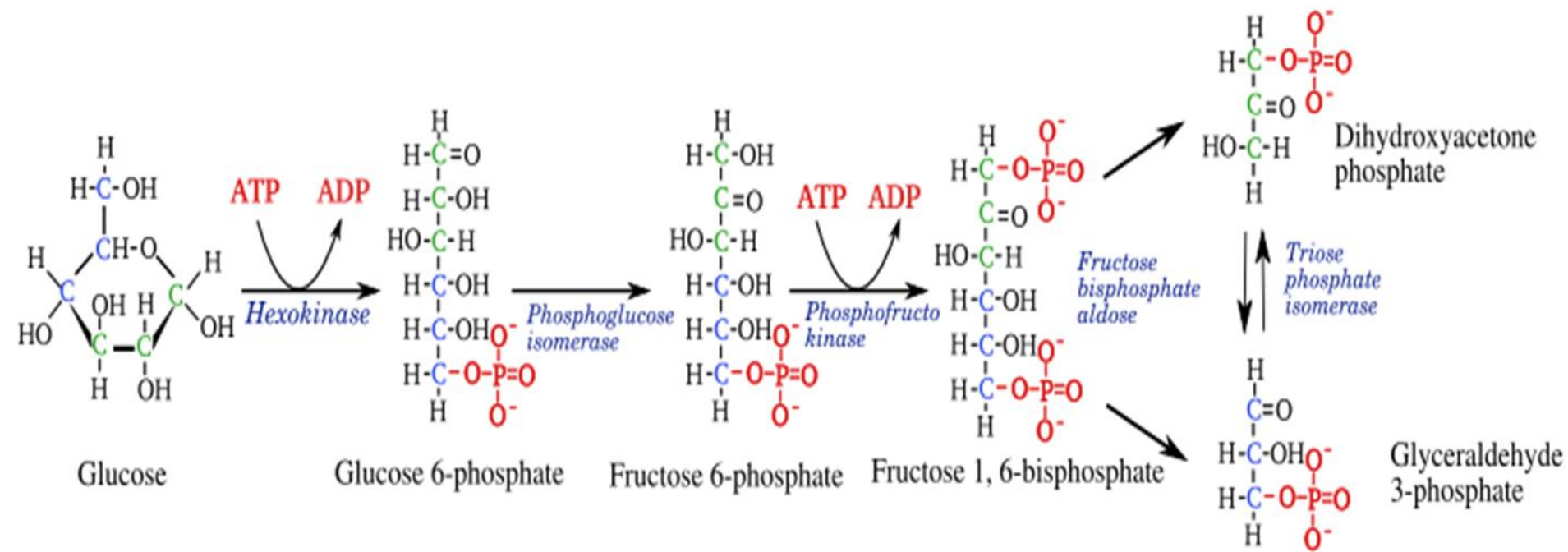
- ✓ Major pathways: A. Glycolysis B. Krebs' cycle
- ✓ Minor pathways: A. Pentose shunt B. Uronic acid pathway

II. Anabolic fates:

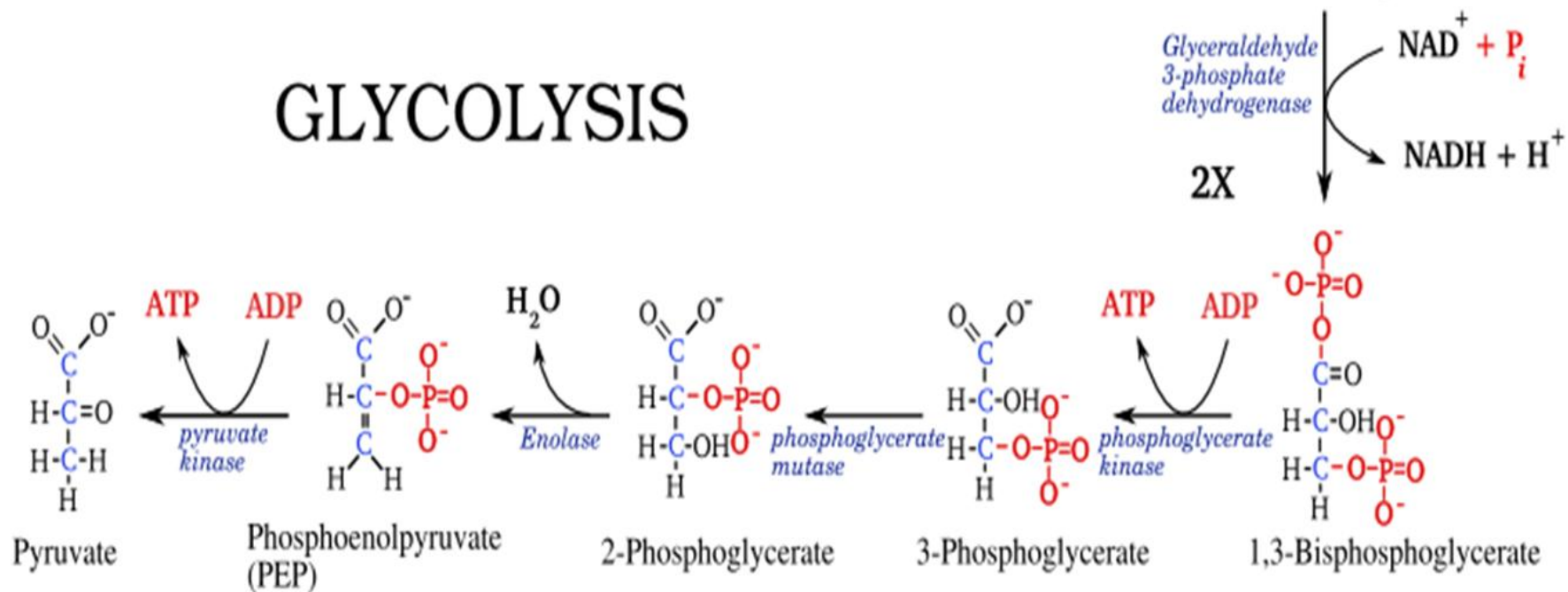
- ✓ Glycogenesis
- ✓ Glycolipids, glycoproteins & Proteoglycans synthesis

Glycolysis (**Embden -Meyerhoff pathway**)

- Glycolysis contain **series of ten enzymatic reactions**
 - breaks down **glucose** into **2 pyruvic acids**
 - occurs in **cytosol**
 - **anaerobic phase** of cellular respiration
 - ✓ yields two ATP molecules per glucose
 - **Aerobic phase** of cellular respiration
 - ✓ Yields 8 ATP molecules per glucose



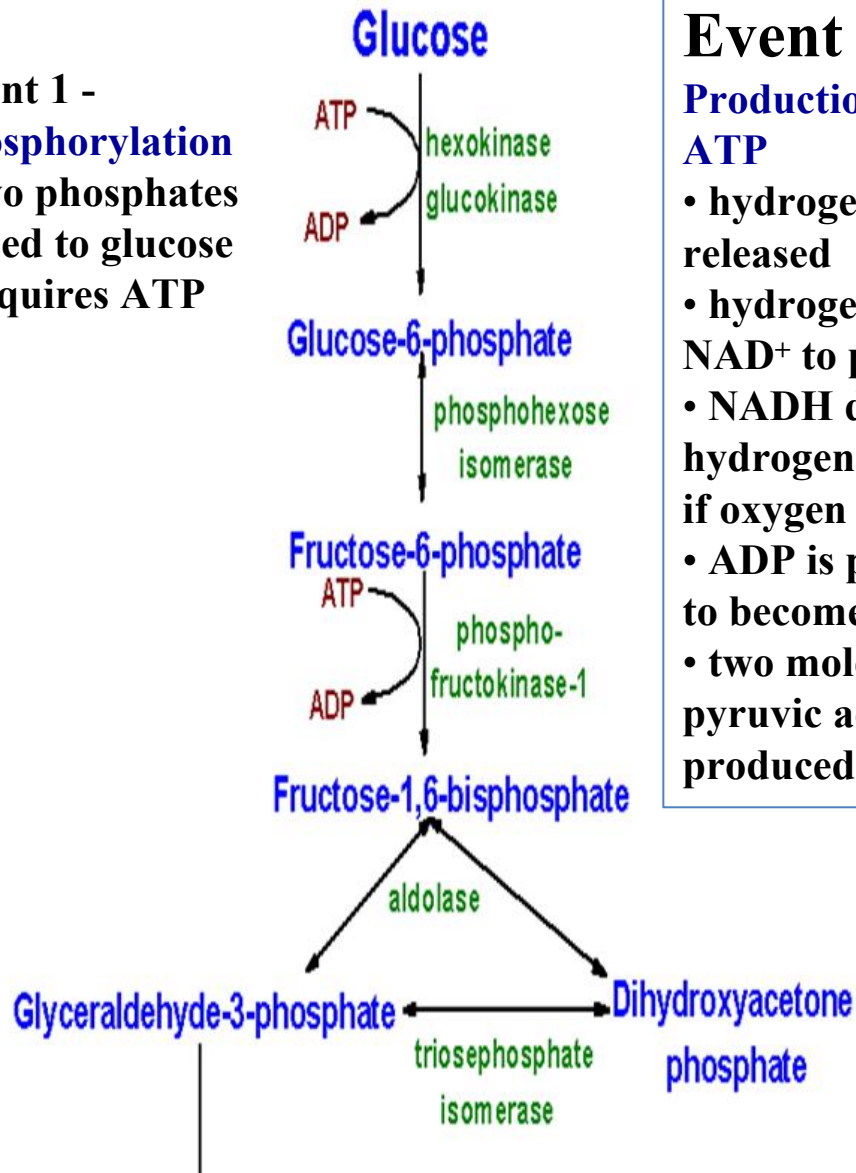
GLYCOLYSIS



Glycolysis (**Embden -Meyerhoff pathway**)

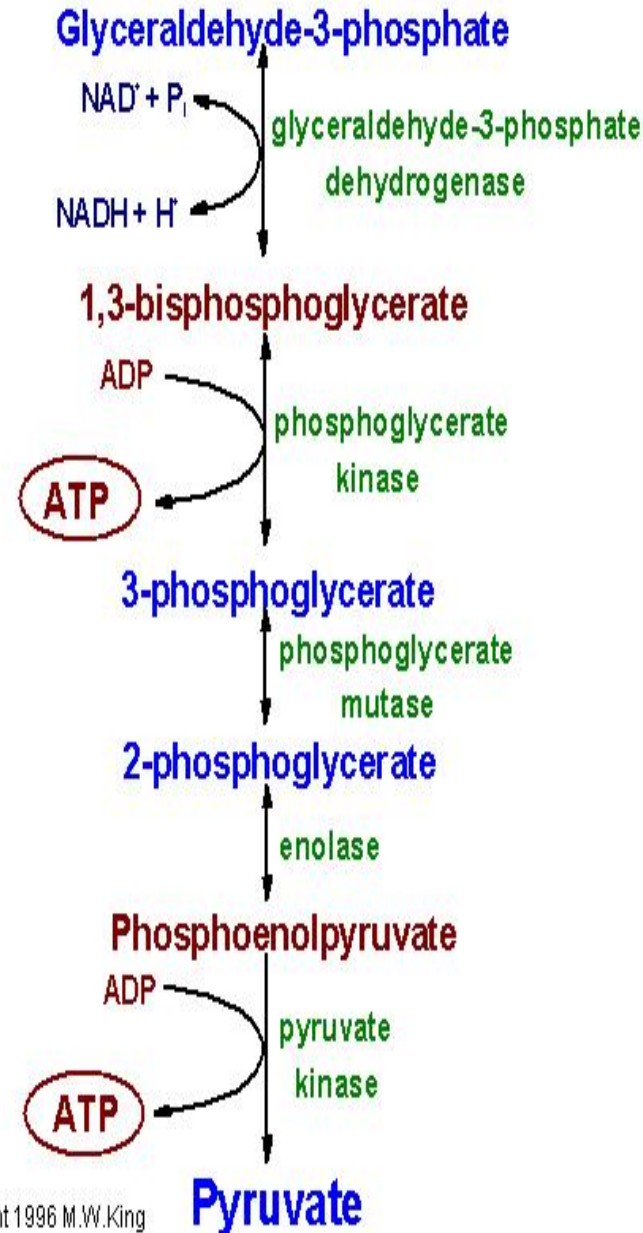
Event 1 - Phosphorylation

- two phosphates added to glucose
- requires ATP



Event 3 – Production of NADH and ATP

- hydrogen atoms are released
- hydrogen atoms bind to NAD^+ to produce **NADH**
- **NADH** delivers hydrogen atoms to ETC if oxygen is available
- **ADP** is phosphorylated to become **ATP**
- two molecules of pyruvic acid are produced



Event 2 – Splitting (cleavage)

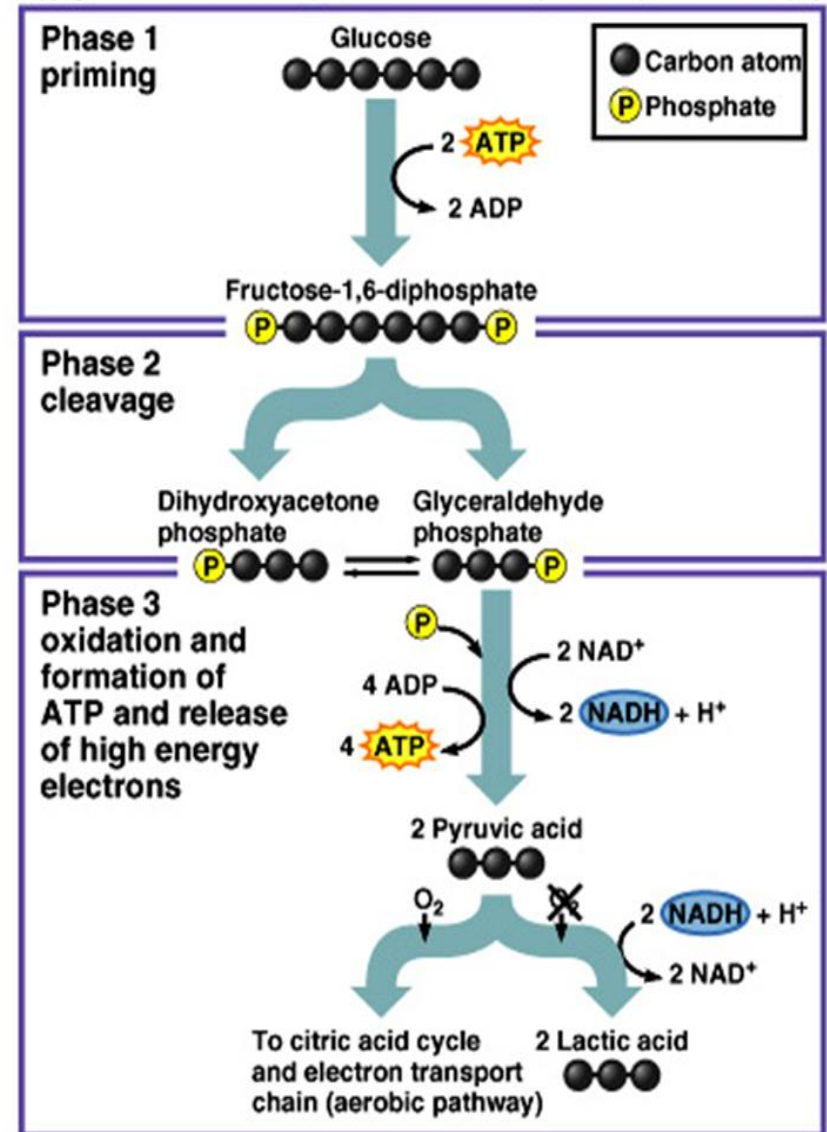
6-carbon glucose split into two 3-carbon molecules

Anaerobic Reactions (Absence of Oxygen)

If oxygen is not available -

- electron transport chain cannot accept NADH
- pyruvic acid is converted to lactic acid
- glycolysis is inhibited
- ATP production declines

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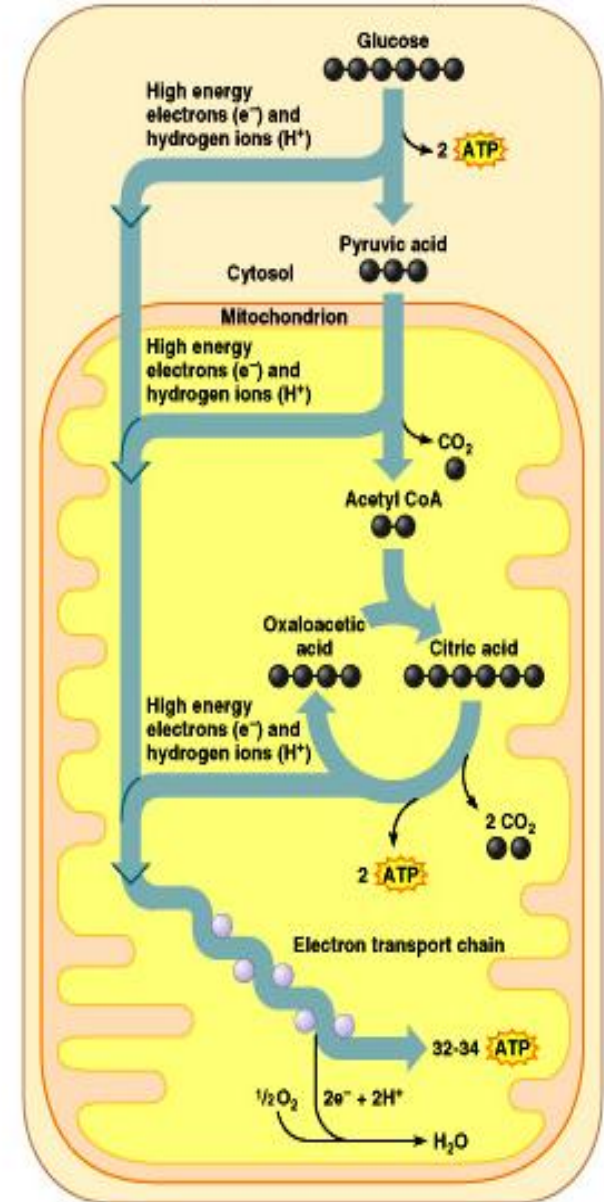


Aerobic Reactions (Presence of Oxygen)

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If oxygen is available –

- pyruvic acid is used to produce **acetyl CoA**
- **citric acid cycle** begins
- ETC functions
- CO_2 and H_2O are formed
- 38 ATP / per glucose



Differences between aerobic and anaerobic glycolysis:

	Aerobic	Anaerobic
1. End product	Pyruvate	Lactate
2 .energy	8 ATP	2 ATP
3.Regenerationof NAD^+	Through respiration chain in mitochondria	Through Lactate formation
4. Availability to TCA in mitochondria	Available and 2 Pyruvate can oxidize to give 30 ATP	Not available as lactate is cytoplasmic substrate

Importance of lactate production in anaerobic glycolysis:

- In absence of oxygen, lactate is the end product of glycolysis:

Glucose → Pyruvate → Lactate

- In absence of oxygen, $\text{NADH} + \text{H}^+$ is not oxidized by the respiratory chain.
- The conversion of pyruvate to lactate is the mechanism for regeneration of NAD^+ .
- This helps continuity of glycolysis, as the generated NAD^+ will be used once **more for oxidation of another glucose molecule.**

Coris Cycle or Lactic Acid Cycle

- In an actively contracting muscle, only about 8% of the pyruvate is utilized by TCA cycle and
- The remaining reduced to lactate.
 - The lactic acid thus generated
 - ✓ accumulate in the muscle tissues.
 - ✓ Causes the **muscle cramps**, often associated with **strenuous muscular exercise**
- During exercise, blood lactate level increases considerably
 - Lactate diffuse into blood
 - then reaches **liver** where it is **oxidized to pyruvate**.
 - It is then taken up through **gluconeogenesis pathway** and becomes glucose
 - The glucose enter into blood and then taken to muscle.
 - This cycle is called **cori's cycle**,

Significance of the cycle

- by which the lactate is **efficiently reutilized by the body**
- Muscle cannot form glucose by gluconeogenesis process because
 - glucose 6 phosphatase is absent.
- Unlike Liver, muscle **cannot supply Glucose** to other organs in spite of having Glycogen

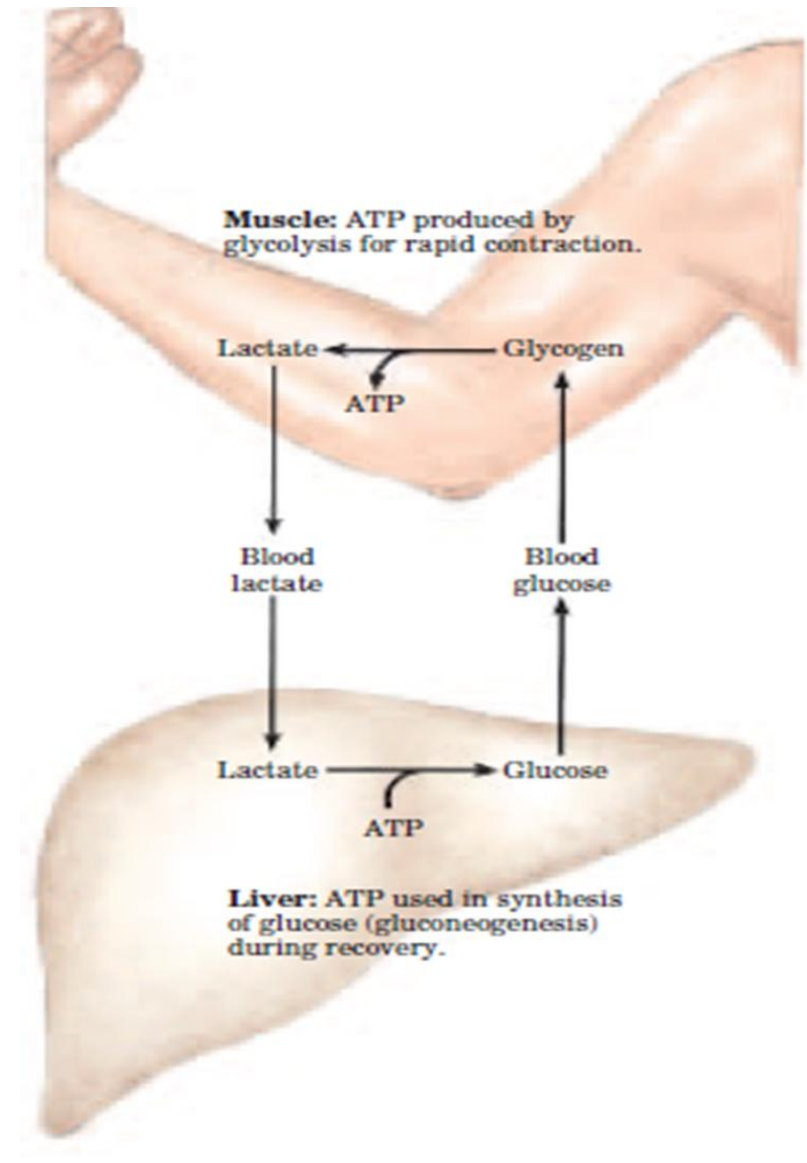
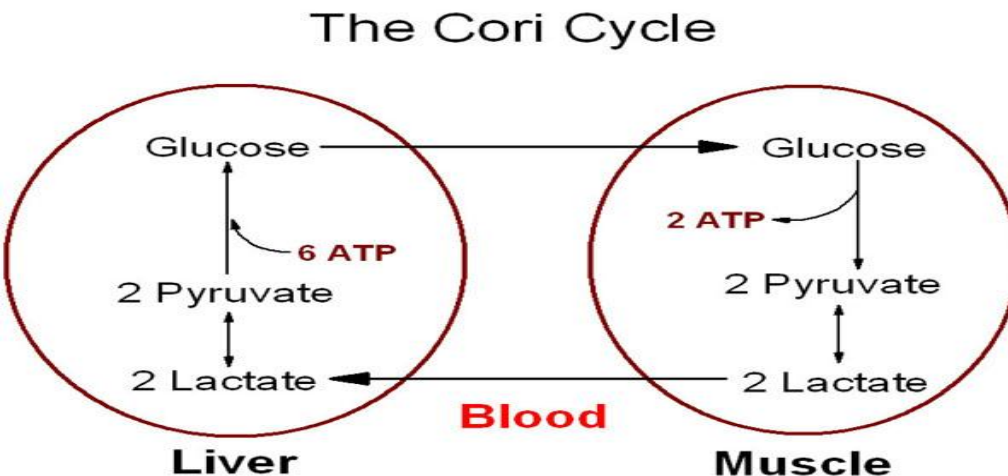


Fig . The Cori cycle.

Glycogen Metabolism In Muscle

I. Glycogenesis

Glycogenesis(Glycogen synthesis)

- ✓ Glycogen synthesis takes place in virtually all animal tissues but is especially prominent in the liver and skeletal muscles.
- ❖ Carbohydrates are stored mainly in the form of **glycogen** for the following reasons:
 - ✓ **Dietary intake of glucose** and **glucose precursors** are sporadic.
 - ✓ **Storage of glucose** in the form of **fat is not suitable** because:
 - **Fat cannot be mobilized** as rapidly in muscles as glycogen.
 - **Fat cannot be used as a source of energy** in the absence of oxygen.
 - Unlike glycogen fat **cannot be transformed directly to glucose**

I. Glycogenesis

Glycogen Function

❖ In muscle

- The **synthesis** and **breakdown** of glycogen regulated
 - To meet the energy requirements of the muscle cell.

Sources of Glucose for glycogenesis in muscle

- **Blood glucose only.**

Site of glycogenesis:- **cytoplasm**

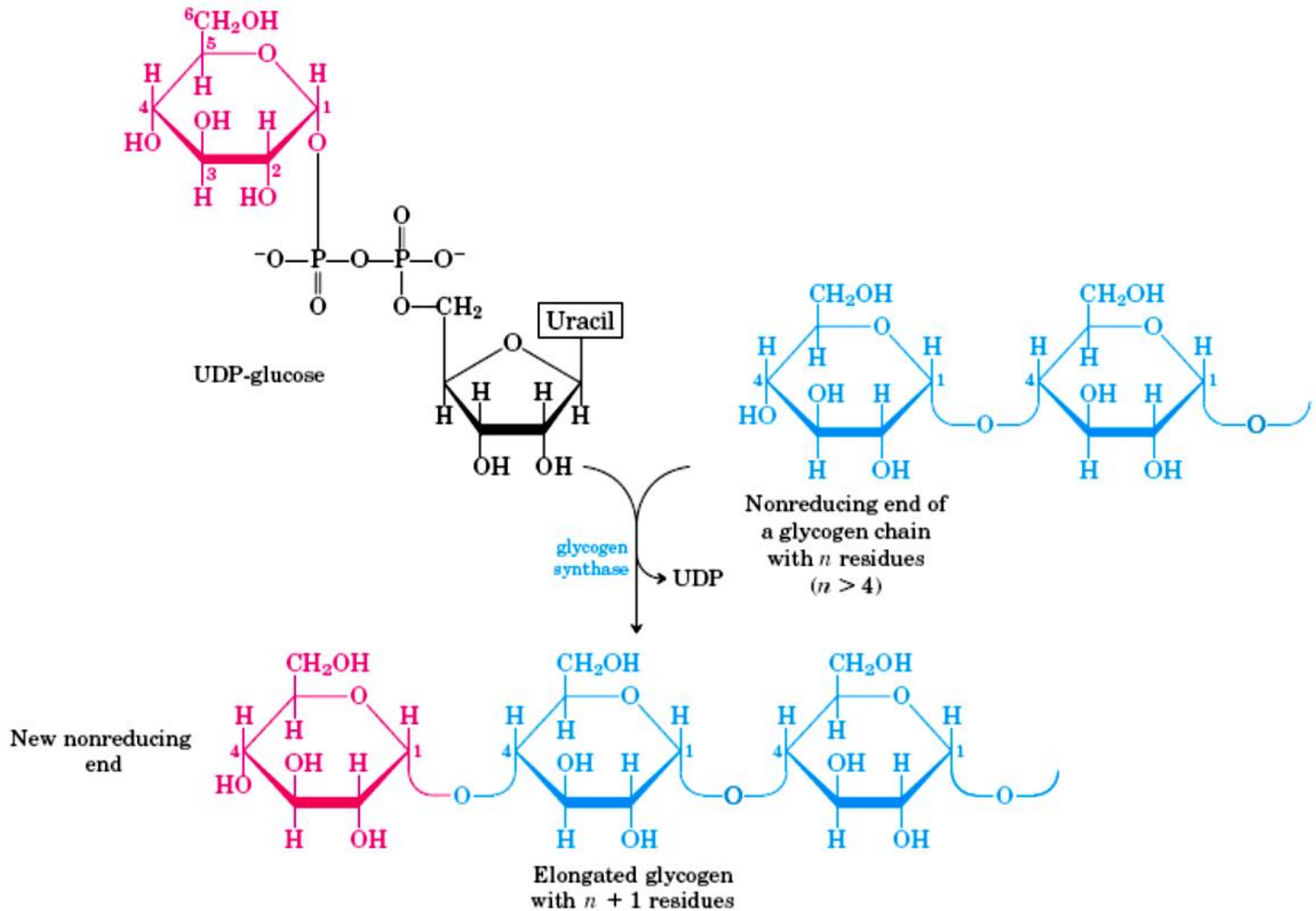
☞ **Steps:**

- The starting point for synthesis of glycogen is **glucose 6-phosphate**

I. Glycogenesis

- **Step1; Glucose 6-phosphate;** It convert to **glucose 1-phosphate** by **phosphoglucomutase**
$$\text{Glucose 6-phosphate} \rightleftharpoons \text{glucose 1-phosphate}$$
- **Step2 ; Glu-1-p**  **UDP-glucose(active nucleotide)** by the following reaction
- $$\text{Glucose 1-phosphate} + \text{UTP} \longrightarrow \text{UDP-glucose} + \text{PP}_i$$
- Catalyzed by **UDP-glucose pyrophosphorylase**
- The sugar nucleotide **UDP-Glucose** donates glucose for **glycogen synthesis**
- **Step3; connects C₁ of glucose** units of UDP-glucose with **C₄ of a terminal glucose residue** in a pre-existing glycogen chain (**glycogen primer**) by enzyme **Glycogen synthase**
- **Step4; Branching**

I. Glycogenesis



I. Glycogenesis

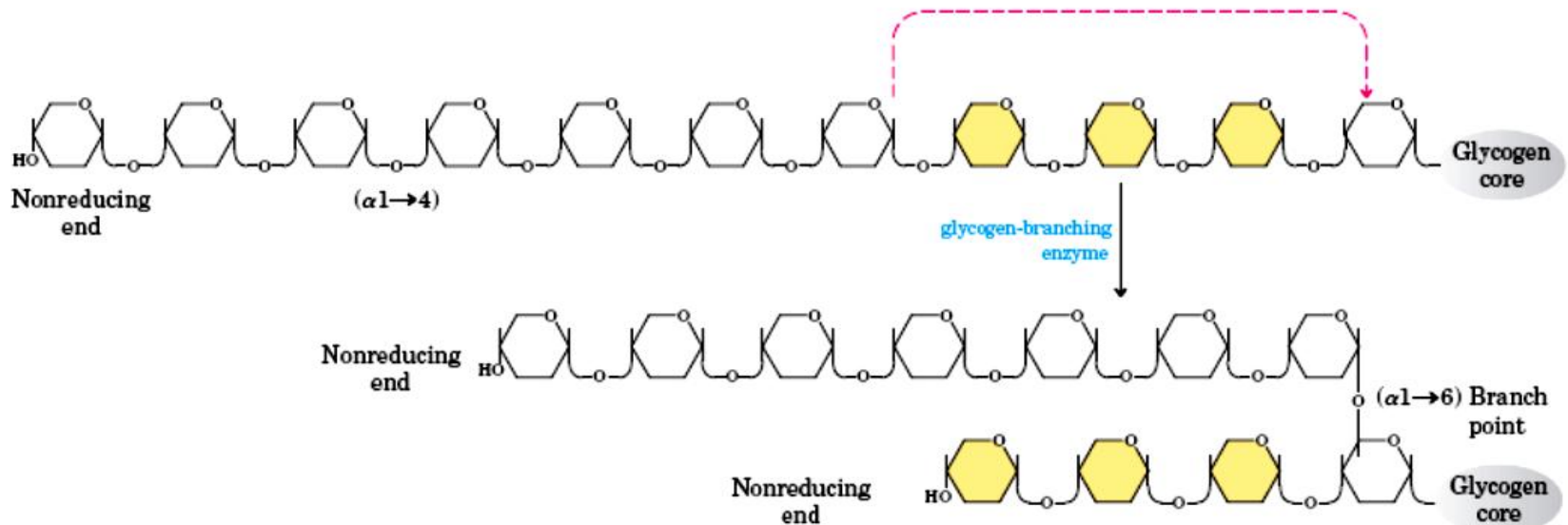
➤ Step4; Branching

✓ The **glycogen-branching enzyme**, also called

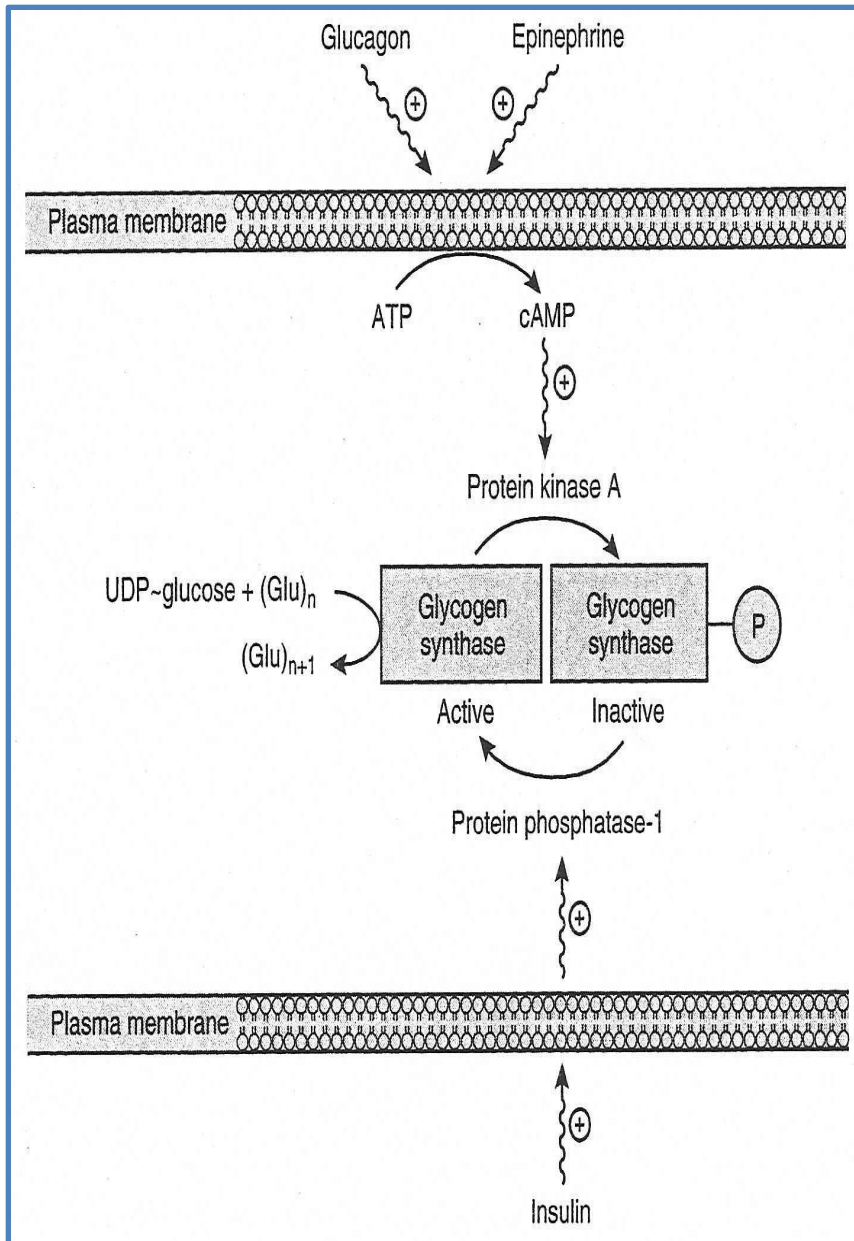
➤ **amylo (1→4) to (1→6) transglycosylase** or

➤ **Glycosyl-(4:6)-Transferase:**

➤ **transfer of 6 or 7 glucose residues** from the nonreducing end of a glycogen branch having **at least 11** residues to the C₆ hydroxyl group of a glucose residue



Regulation of Glycogenesis



The key regulatory enzyme of glycogenesis

glycogen synthase.

- exists in 2 forms:
- Inactive form**
 - glycogen synthase b**
 - This form is phosphorylated, i.e., phospho-synthase.
 - By protein kinase
- Active form**
 - glycogen synthase a**
 - Dephospho-synthase results from dephosphorylation of synthase
 - by phosphatase.

II. Glycogenolysis

➤ Glycogen Breakdown(Glycogenolysis) is process of **glycogen catabolism** into:

- **Glucose** ➡ released to blood, in the *liver* Or
- **Glucose-6-phosphate** ➡ **Lactic acid**, in the *skeletal muscles*.

➤ through the action of **three enzymes**:

- ✓ **Glycogen phosphorylase,**
- ✓ **Glycogen debranching enzyme, &**
- ✓ **Phosphoglucomutase.**

➤ **Site**

- ✓ Cytoplasm of **all cells**
- ✓ but high activity in ***liver and muscles***

II. Glycogenolysis

Steps:

1. Glycogen phosphorylase

- catalyzes the reaction ($\alpha 1 \rightarrow 4$) glycosidic linkage
- removing the terminal glucose residue as
 - ✓ **α -D-glucose 1-phosphate**
- **Pyridoxal phosphate (PLP)** is an essential cofactor in the reaction
 - ✓ its phosphate group acts on the glycosidic bond.
- ***acts repetitively*** on the **non reducing ends**
 - until it reaches a point **four glucose residues** away from an **($\alpha 1 \rightarrow 6$) branch point**

II. Glycogenolysis

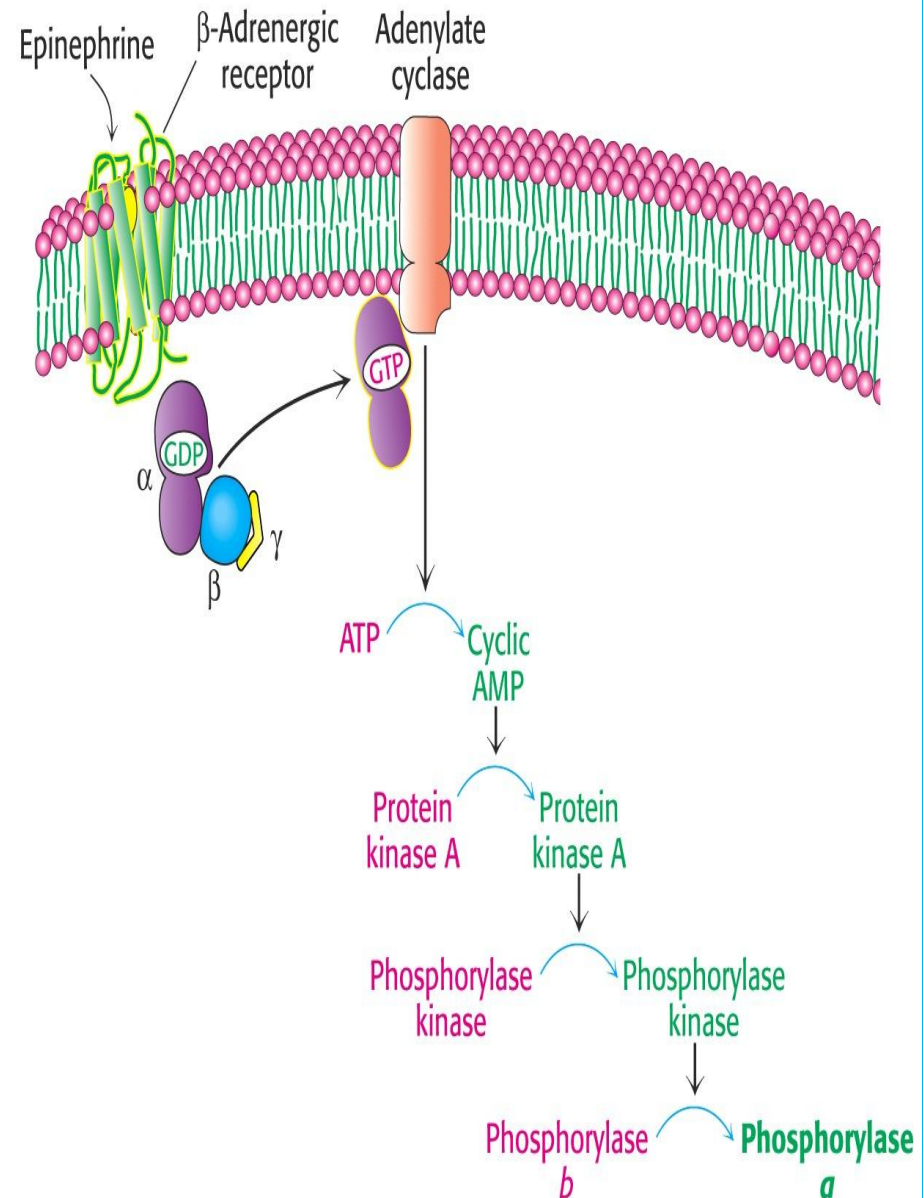
2. Further degradation by **glycogen phosphorylase**

- can occur only after the **debranching enzyme**
- also known as **oligo ($\alpha 1 \rightarrow 6$) to ($\alpha 1 \rightarrow 4$) glucantransferase**
 - catalyzes two successive reactions
 - **First, the *transferase activity***
 - the enzyme shifts **three glucose residues** from the branch to a near by nonreducing end
 - **Second ,cutting the glucose remaining** at the branch point ($\alpha 1 \rightarrow 6$)
 - then **released as free glucose** by
 - the enzyme's **($\alpha 1 \rightarrow 6$) glucosidase activity**.
 - Then **glycogen phosphorylase** activity continue.



Regulation of Glycogenolysis

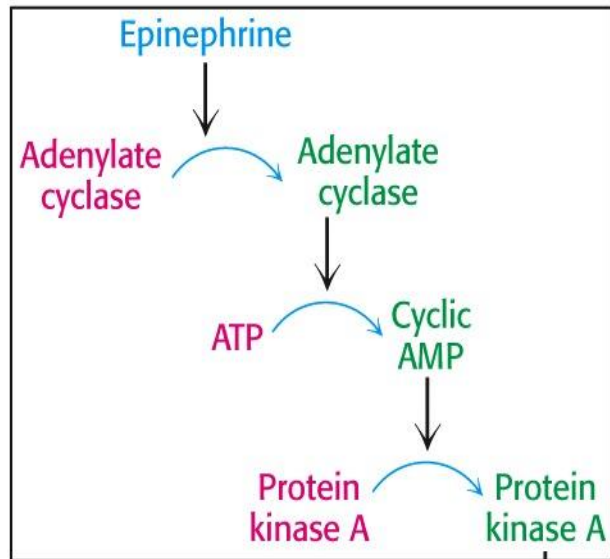
- ❖ The key regulatory enzyme
 - **Glycogen phosphorylase.**
 - present in **2 forms**:
 - **Active**
 - ✓ **Phosphorylase a**
 - ✓ **phosphorylase form.**
 - **Inactive**
 - ✓ **phosphorylase b**
 - ✓ **dephosphorylase form.**



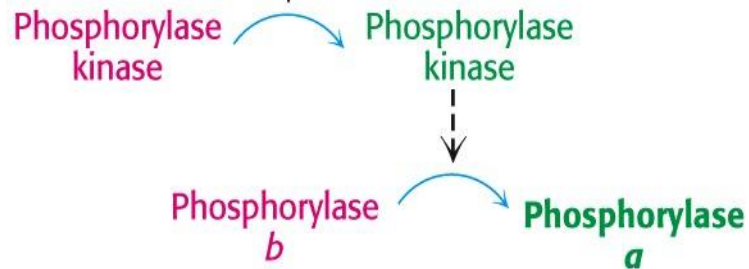
Regulation of Glycogen Metabolism

Glycogen synthesis and breakdown are reciprocally regulated

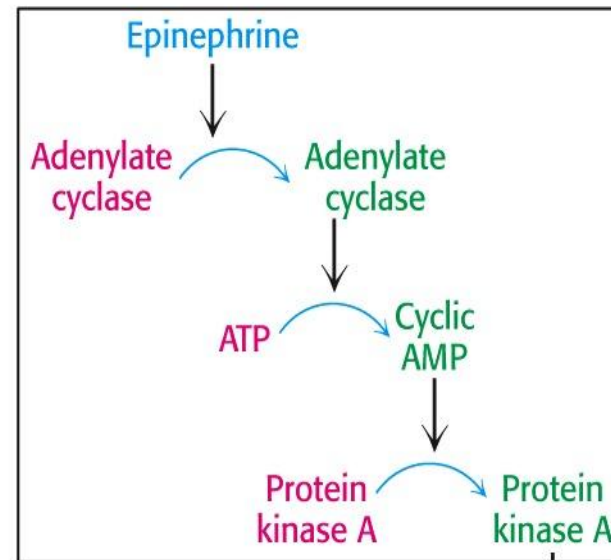
Red=inactive forms, green = active forms.



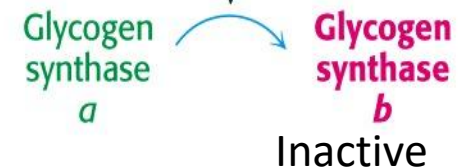
(A)



Active



(B)



Inactive

✓ Protein phosphatase 1 (PP1) regulates glycogen metabolism via Insulin activation.

ANY QUESTIONS?

THANK you